

Python Lists – Notes & Code Examples

A beginner-friendly guide to Python List concepts and important programs.

1. Create a List

```
numbers = [10, 20, 30, 40, 50]
print(numbers)
```

2. Access List Elements

```
numbers = [10, 20, 30, 40, 50]

print(numbers[0])
print(numbers[2])
print(numbers[-1])
```

3. Change an Element

```
numbers = [10, 20, 30, 40]

numbers[1] = 25
print(numbers)
```

4. Add Elements

```
numbers = [10, 20, 30]

numbers.append(40)
numbers.insert(1, 15)

print(numbers)
```

5. Remove Elements

```
numbers = [10, 20, 30, 40]

numbers.remove(20)
numbers.pop()

print(numbers)
```

6. List Slicing

```
numbers = [10, 20, 30, 40, 50]

print(numbers[1:4])
print(numbers[:3])
print(numbers[2:])
print(numbers[::-1])
```

7. Find List Length

```
numbers = [10, 20, 30, 40]

print(len(numbers))
```

8. Loop Through a List

```
numbers = [10, 20, 30, 40]

for num in numbers:
    print(num)
```

9. Find Largest and Smallest

```
numbers = [10, 25, 5, 40, 15]

print(max(numbers))
print(min(numbers))
```

10. Sort a List

```
numbers = [40, 10, 30, 20]

numbers.sort()
print(numbers)
```

11. Reverse a List

```
numbers = [10, 20, 30, 40]

numbers.reverse()
print(numbers)
```

12. List Comprehension

```
numbers = [1, 2, 3, 4, 5]

squares = [x * x for x in numbers]
print(squares)
```

13. Get Even Numbers

```
numbers = [1, 2, 3, 4, 5, 6, 7, 8]

even = [x for x in numbers if x % 2 == 0]
print(even)
```

14. Count an Element

```
numbers = [10, 20, 10, 30, 10]

print(numbers.count(10))
```

15. Find an Element's Index

```
numbers = [10, 20, 30, 40]

print(numbers.index(30))
```

16. Extend a List

```
a = [1, 2, 3]
b = [4, 5, 6]

a.extend(b)
print(a)
```

Important Python List Methods

`append()` - adds one element
`insert()` - adds an element at a specific position
`extend()` - adds multiple elements
`remove()` - removes a specified element
`pop()` - removes an element by index / last element
`clear()` - removes all elements
`index()` - returns the position of an element
`count()` - counts occurrences
`sort()` - sorts the list
`reverse()` - reverses the list
`copy()` - creates a copy of the list

Practice Questions

1. Create a list of 10 numbers and print all elements.
2. Find the sum of all elements in a list.
3. Find the largest number without using `max()`.
4. Find the smallest number without using `min()`.
5. Count how many even numbers are in a list.
6. Reverse a list without using `reverse()`.
7. Remove duplicate elements from a list.
8. Find the second largest number.
9. Create a list of squares from 1 to 10.
10. Separate even and odd numbers into two lists.